

CBSE Class 12 Physics — Important Derivations

A chapter-wise checklist of the derivations most likely to appear in the CBSE Class 12 Physics board exam — the ones that repeat, year after year, as 2-mark, 3-mark, or 5-mark questions. This is a **list to work through**, not the derivations themselves — the idea is to use it as a revision tracker: can you reproduce each one on paper, from the starting principle, without looking it up?

Note on currency: CBSE revises (rationalises) the Physics syllabus most years, and has dropped several derivations in recent cycles — Cyclotron, Velocity Selector, the Toroid, the magnetic dipole moment of a revolving electron, Earth's Magnetism, Eddy Currents, and Brewster's Law are **not** in this list because they were removed from the syllabus in past rationalisations. This version has also been cross-checked against the current official syllabus wording for which topics are explicitly marked "*no derivation, formula only*" or "*qualitative treatment only*" — those are called out separately below rather than listed as derivations. Still, cross-check against the current year's official CBSE curriculum (cbseacademic.nic.in) before finalising a revision plan, since the exact wording can shift year to year.

Frequency key: ★★★ = asked almost every year, in some form · ★★ = asked frequently · ★ = asked occasionally, still worth knowing.

Status tags used below: a plain table row is a full derivation you should be able to reproduce.

Formula only means the syllabus explicitly says no derivation is required — know and apply the formula, don't derive it. **Qualitative only** means no maths is expected at all — understand the idea and be able to explain it in words/diagram.

Unit 1 — Electrostatics

Derivation	Typical marks	Frequency
Electric field due to a point charge (from Coulomb's law)	2	★★
Electric field on the axial line of an electric dipole	3	★★★
Electric field on the equatorial line of an electric dipole	3	★★★
Torque on an electric dipole in a uniform external electric field	2–3	★★★
Electric field due to an infinite plane sheet of charge (Gauss's law)	3	★★★
Electric field due to a uniformly charged thin spherical shell — inside, on, and outside the surface (Gauss's law)	3–5	★★★

Derivation	Typical marks	Frequency
Electric field due to an infinitely long straight charged wire (Gauss's law)	3	★★
Electric potential due to a point charge	2	★★
Electric potential due to an electric dipole, at a general point	3	★★
Relation between electric field and potential ($E = -dV/dr$)	2	★★
Capacitance of a parallel plate capacitor (no dielectric)	3	★★★★
Capacitance of a parallel plate capacitor with a dielectric slab	3–5	★★★★
Effective capacitance of capacitors in series and in parallel	2–3	★★

Formula only: Energy stored in a capacitor — $U = \frac{1}{2}CV^2 = \frac{1}{2}QV = Q^2/2C$. The syllabus explicitly says "*no derivation, formulae only*" — know the three equivalent forms and when to use each, don't derive it.

Unit 2 — Current Electricity

Derivation	Typical marks	Frequency
Relation between current and drift velocity ($I = nAeV_d$)	3	★★★★
Ohm's law from the microscopic (drift velocity) model; resistivity $\rho = m/(ne^2\tau)$	3–5	★★★★
Balance condition of the Wheatstone bridge	2–3	★★
Principle of a potentiometer — comparing EMFs of two cells / finding internal resistance	3–5	★★

Unit 3 — Moving Charges and Magnetism

Derivation	Typical marks	Frequency
Magnetic field at the centre of a current-carrying circular loop (Biot–Savart law)	3	★★★★
Magnetic field on the axis of a current-carrying circular loop	3–5	★★★★
Magnetic field due to a straight current-carrying conductor (Biot–Savart law)	3	★★
Magnetic field inside a long solenoid (Ampere's circuital law)	3	★★★★

Derivation	Typical marks	Frequency
Force between two long parallel current-carrying conductors; definition of the ampere	3	★★★
Torque on a current-carrying loop in a uniform magnetic field; principle of a moving-coil galvanometer	3–5	★★★
Conversion of a galvanometer into an ammeter (shunt) and a voltmeter (series resistance)	2–3	★★

Qualitative only (Magnetism and Matter): a bar magnet as an equivalent solenoid, and the magnetic field intensity of a dipole on its axial and equatorial lines, are both listed as qualitative treatment only — no derivation is expected for either.

Unit 4 — Electromagnetic Induction & Alternating Current

Derivation	Typical marks	Frequency
Motional EMF (from the force on a moving charge)	3	★★★
Self-inductance of a long solenoid	3	★★★
Mutual inductance of two long coaxial solenoids	3–5	★★
AC generator — derivation of the instantaneous EMF ($e = e_0 \sin \omega t$) from a coil rotating in a magnetic field	3	★★★
Relation between the RMS and peak values of alternating current/voltage ($I_{\text{rms}} = I_0/\sqrt{2}$)	2	★★
AC through a pure resistor / pure inductor / pure capacitor — current, phase relationships	2–3 each	★★
Impedance of a series LCR circuit using a phasor diagram	5	★★★
Resonance condition and resonant frequency of a series LCR circuit	3	★★
Average power in an AC circuit; power factor	2–3	★★
Working and EMF equation of a transformer ; efficiency	3	★★

Unit 5 — Ray Optics & Optical Instruments

Derivation	Typical marks	Frequency
Mirror formula for a concave mirror (real image case)	3	★★
Relation between critical angle and refractive index for total internal reflection ($n = 1/\sin i_c$)	2	★★
Refraction at a single spherical surface — relation between u , v , R	3–5	★★★★
Lens maker's formula	3–5	★★★★
Thin lens formula ($1/v - 1/u = 1/f$)	2–3	★★
Equivalent focal length/power of two thin lenses in contact	2–3	★★
Refraction through a prism — $i + e = A + \delta$, and $n = \sin[(A+D_m)/2] / \sin(A/2)$	3–5	★★★★
Magnifying power of a simple microscope	2–3	★★
Magnifying power of a compound microscope (normal adjustment)	3–5	★★★★
Magnifying power of an astronomical telescope (normal adjustment)	3–5	★★★★

Unit 6 — Wave Optics

Derivation	Typical marks	Frequency
Proof of the laws of reflection and refraction using Huygens' principle	3	★★★★
Conditions for constructive and destructive interference (path difference)	2–3	★★★★

Formula only: Young's double-slit fringe width, $\beta = \lambda D/d$. The syllabus explicitly says "*expression for fringe width (no derivation, final expression only)*" — know what each symbol means and be able to apply it, don't derive it.

Qualitative only: single-slit diffraction (the pattern, and why the central maximum is twice as wide as the secondary ones) — a mathematical derivation of the fringe/angular width is not expected, only the qualitative explanation.

Unit 7 — Dual Nature of Radiation and Matter

Derivation	Typical marks	Frequency
Einstein's photoelectric equation	3	★★★★
de Broglie wavelength of a moving particle	2–3	★★★★

Unit 8 — Atoms

Derivation	Typical marks	Frequency
Bohr's postulates → radius of the nth orbit of the hydrogen atom	3–5	★★★
Bohr's model → velocity of the electron in the nth orbit	2–3	★★
Bohr's model → total energy of the electron in the nth orbit	3–5	★★★

Qualitative only: the hydrogen line spectra (Lyman, Balmer, Paschen series, etc.) and the Rydberg formula are listed as qualitative treatment only — know the series and what the formula predicts, but a derivation of the formula itself isn't expected.

Unit 9 — Nuclei

No standalone derivations in the current syllabus. Chapter 13 (Nuclei) now covers only the composition and size of the nucleus, nuclear forces, mass defect, the binding-energy-per-nucleon curve, and fission/fusion — all conceptual, not derivation-based. The radioactivity topic (decay law $N = N_0 e^{(-\lambda t)}$, half-life, and mean life) has been **removed from the syllabus entirely** and should not be prepared as a derivation, or at all, unless your school's internal notes say otherwise.

Unit 10 — Semiconductor Electronics

Mostly conceptual in the current syllabus (V–I characteristics, working principles, logic gates) — no major standalone derivations are typically asked from this unit, unlike Units 1–9 above.

How to use this list

1. Go unit by unit. For each derivation, close the book and try to produce it from the starting principle (Coulomb's law, Gauss's law, Biot–Savart law, Snell's law, Bohr's postulates, etc.) through to the final result, with the diagram.
2. Mark anything you couldn't do cleanly — those go on a short list for a second pass closer to the exam.
3. The ★★★ items are worth over-preparing: they show up almost every year, sometimes as the derivation itself, sometimes as a numerical built on top of it.
4. Board papers often ask you to **derive, then apply** (e.g. derive the lens maker's formula, then use it in a numerical) — so pair each derivation with a couple of numericals on the same formula while revising.

Sources consulted for frequency/scope: recent-year CBSE sample papers and previous-year question papers, and the current rationalised CBSE Class 12 Physics syllabus (cbseacademic.nic.in) — including its exact wording on which items are formula-only or qualitative-only, which is why those are now called out separately rather than listed as derivations.